

AI CAREER PULSE

Decoding the Job Market

A Tableau-Based Business Intelligence Deep Dive into the 2025–2026 AI Job Market

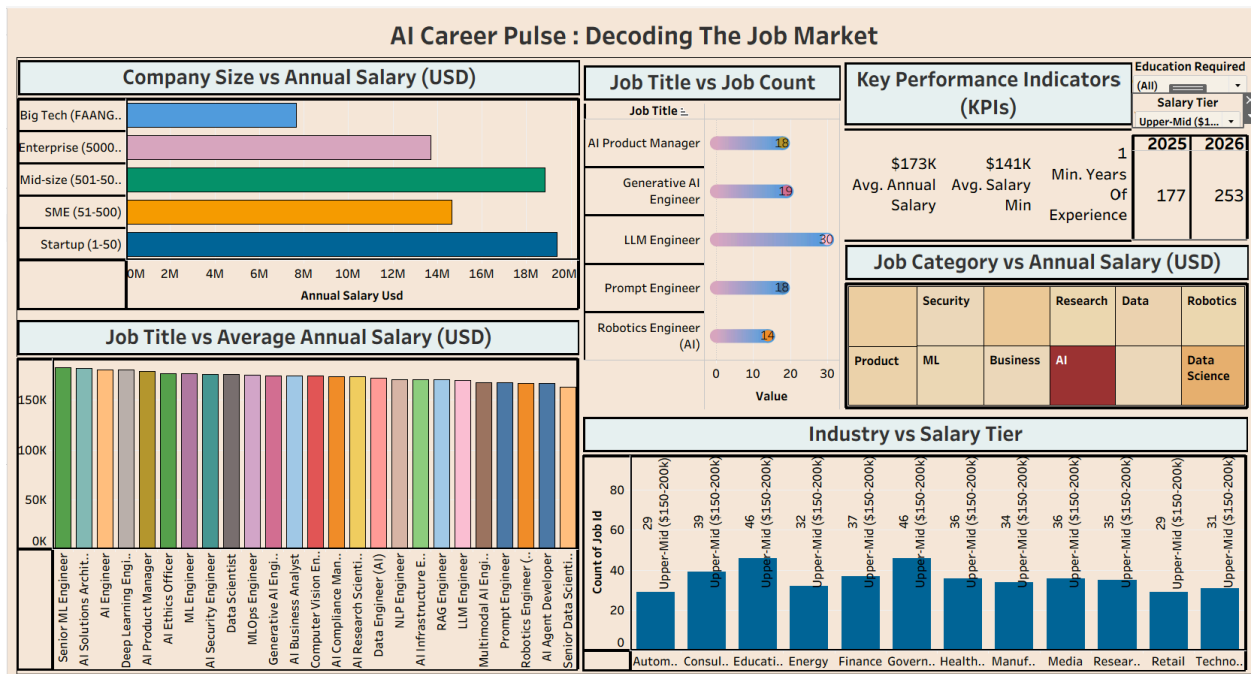
Field	Detail
Project Title	AI Career Pulse: Decoding the Job Market
Prepared By	Aman Srivastava
Program	SkillWallet Data Analytics Program
Tool Used	Tableau (Desktop / Public)
Dataset	ai_jobs_market_2025_2026 (custom AI job postings dataset)
GitHub Repository	Aman-2305/ai-career-pulse
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Date	[Insert Submission Date]

1. Executive Summary

AI Career Pulse is an end-to-end business intelligence project built in Tableau to analyze the 2025–2026 AI job market. The project ingests a curated dataset of AI-related job postings — spanning job titles, categories, industries, company sizes, salary bands, experience levels, and required education — and transforms it into a set of interactive dashboards and a guided data story.

The final deliverable consists of 8 supporting worksheets, 1 consolidated interactive dashboard, and a 5-page narrative story, all cross-filtered by Country, Education Required, and Salary Tier. The analysis is designed to serve three distinct audiences: job seekers deciding which AI skill to invest in, recruiters benchmarking compensation, and industry analysts tracking macro hiring trends.

Headline findings include: AI Engineering is by far the most valuable job category by total salary pool; company size has surprisingly little correlation with pay; Consulting and Government are emerging as major (and underrated) AI employers; and — most notably — average salary is nearly flat across experience levels, suggesting the market currently rewards AI skill possession more than tenure.



2. Problem Statement

2.1 Background

The artificial intelligence job market has grown at an unprecedented pace between 2025 and 2026, with new role types (e.g., Prompt Engineer, AI Agent Developer, Generative AI Engineer) emerging faster than standard compensation benchmarks can track them. Information about these roles is fragmented across job boards, salary-survey PDFs, and anecdotal social media posts, making it difficult for any single stakeholder to form an accurate, data-backed picture of the market.

2.2 Core Problem

There is no single, interactive, filterable view that lets a user answer practical questions such as:

- Which AI job category or title pays the most, on average and in total?
- Does company size (Startup vs. Big Tech) meaningfully change compensation?
- Which industries are hiring most aggressively for AI talent, beyond the obvious "Technology" sector?
- Does more experience actually translate into proportionally higher pay in AI roles?
- How is hiring volume trending year-over-year (2025 → 2026)?

2.3 Objective of This Project

To design and build a Tableau dashboard and story that consolidates AI job market data into clear, interactive, and filterable visualizations — enabling data-driven decisions for job seekers, recruiters, and analysts without requiring any manual spreadsheet analysis.

2.4 Target Stakeholders

Stakeholder	Key Question Answered
Job Seeker	Which AI role and skill set offers the best pay-to-entry-barrier ratio?
Recruiter / HR	What is a competitive salary band to offer for a given role, industry, and company size?
Industry Analyst	Where is AI hiring demand concentrated, and how is it shifting year over year?

3. Project Objectives

1. Consolidate raw AI job-posting data into a single, clean Tableau data source.
2. Engineer calculated fields that convert raw numeric/text data into decision-ready metrics (e.g., salary tiers, demand scores, role-type flags).

3. Build a set of focused worksheets, each answering one specific business question.
4. Combine these worksheets into one interactive, filterable dashboard.
5. Construct a 5-page guided Story that walks a non-technical viewer through the same insights in a logical narrative order.
6. Extract and document clear, quantified insights that are useful to each of the three target stakeholders.

4. Dataset Overview

The primary data source is a single flat table, `ai_jobs_market_2025_2026`, connected directly inside Tableau (visible in the Data pane as “`ai_jobs_market_2025...`”).

4.1 Dataset Scale

- Total job postings analyzed: 91
- Postings by year: 2025 → 36 postings, 2026 → 55 postings (a ~53% year-over-year increase)
- Distinct job titles: 23 (e.g., Senior ML Engineer, AI Solutions Architect, AI Research Scientist, Prompt Engineer, Generative AI Engineer, Robotics Engineer (AI), Computer Vision Engineer, etc.)
- Job categories: 11 (Architecture, Governance, Business, Security, Research, ML Operations, Data Engineering, Robotics, Product, AI Engineering, Data Science)
- Industries covered: 12 (Automotive, Consulting, Education, Energy, Finance, Government, Healthcare, Manufacturing, Media, Research, Retail, Technology)
- Company size bands: 5 (Startup 1–50, SME 51–500, Mid-size 501–5000, Enterprise 5000+, Big Tech / FAANG+)
- Experience levels: 4 (Entry 0–2 yrs, Mid 3–5 yrs, Senior 6–9 yrs, Lead 10+ yrs)

4.2 Raw Fields (as found in the Data pane)

Field Name	Description
Job Id	Unique identifier per job posting (used with COUNTD for distinct counts)
Job Title	Specific role name, e.g., "Senior ML Engineer", "Prompt Engineer"
Required Skills	Text field listing skills required for the posting
Posting Month / Posting Year	Temporal fields used for trend analysis
Remote Work	Indicates remote / hybrid / on-site status
Salary Tier	Pre-binned salary band, e.g., "Upper-Mid (\$150–200k)"
Annual Salary USD	Full annual compensation in USD
Salary Min USD / Salary Max USD	Lower and upper bound of the posted salary range
Years Of Experience	Numeric years of experience required
Benefits Score 10	A normalized 0–10 score representing benefits generosity
Demand Score	A normalized score representing hiring demand intensity for the role
Demand Growth YoY	Year-over-year percentage growth in demand for the role
Is Llm Role / Is Remote Friendly / Is Senior	Boolean/flag fields for role classification
Latitude / Longitude (generated)	Auto-generated geographic fields (for potential map visuals)

5. Data Preparation & Preprocessing

Before building any visualization, the raw dataset was reviewed and enriched with calculated fields inside Tableau so that every downstream worksheet could reuse consistent, decision-ready metrics rather than raw numbers.

5.1 Data Cleaning Steps

- Verified there were no duplicate Job Id values that would inflate COUNTD-based counts.
- Standardized categorical text fields (Job Category, Industry, Company Size) to consistent casing/spelling so that Tableau does not split a single real-world category into multiple bars due to typos.
- Confirmed numeric fields (Annual Salary USD, Salary Min USD, Salary Max USD, Years Of Experience) were correctly typed as numbers, not strings, so aggregations (SUM/AVG/MIN) compute correctly.
- Reviewed Posting Month / Posting Year for consistent date formatting to support the year-over-year trend view.

5.2 Calculated Fields Created

Calculated Field	Purpose
Avg Salary Range	Derives a representative mid-point / range from Salary Min USD and Salary Max USD for comparison across roles
AI Salary Premium Pct	Calculates the % salary premium an AI-specific role commands versus a baseline, highlighting the AI skill premium
Demand Score / Demand Growth Yoy	Quantifies and tracks how hiring demand for a role is changing over time
Is Llm Role (Count)	Flags whether a posting is for an LLM-specific role, enabling LLM-vs-non-LLM comparisons
Is Remote Friendly	Flags remote-eligible postings for remote-work analysis
Is Senior	Flags postings requiring senior-level experience
Job Id (Count (Distinct))	A COUNTD(Job Id) measure used throughout the dashboard to count unique postings rather than duplicate rows

5.3 Salary Tier Binning

Raw Annual Salary USD values were grouped into a Salary Tier dimension (e.g., "Upper-Mid (\$150–200k)") so that comparisons across industries, company sizes, and job titles could be made at a band level rather than requiring a continuous axis — this makes cross-category comparison in bar charts far more readable.

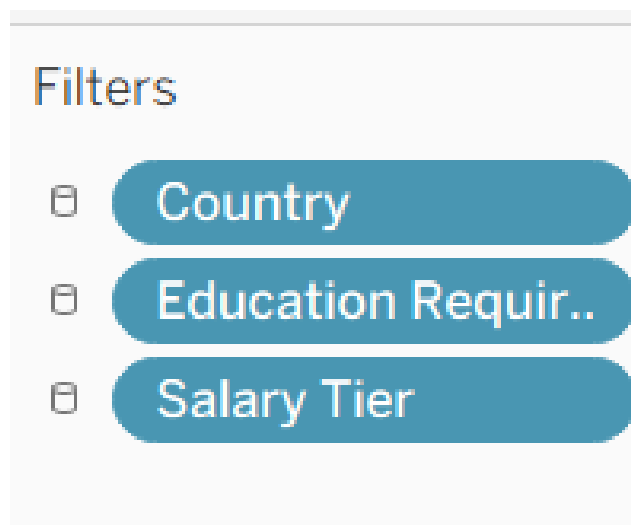
5.4 Filtering Strategy

Three filters were applied globally at the data-source level so that every worksheet (and therefore the dashboard) stays in sync:

- Country — allows drill-down into a specific country or a global (All) view.
- Education Required — allows filtering by minimum qualification (e.g., Associate's, Bachelor's, Master's).
- Salary Tier — allows the viewer to focus on a specific compensation band, e.g., Upper-Mid (\$150–200k), which was used as the primary lens for this analysis.

In addition, Top-N context filters were layered on top of individual views to keep charts focused on the most significant categories:

- Top 5 by COUNTD([Salary Tier]) — restricts a view to the five most common salary tiers.
- Top 5 by COUNTD([Job Id]) — restricts a view to the five job titles/categories with the highest posting counts.



6. Tools & Methodology

6.1 Tools Used

- Tableau Public / Tableau Desktop — for data connection, calculated fields, visualization, dashboarding, and storytelling.
- GitHub — for version control and hosting the project repository (Aman-2305/ai-career-pulse), including data files, workbook, documentation, and screenshots.

6.2 Repository Structure

Folder	Contents
data/	Raw and cleaned dataset(s) used for the analysis
workbook/	The Tableau (.twbx) workbook file
docs/	Project documentation, including this document
assets/screenshots/	Exported screenshots of every worksheet, the dashboard, and the story

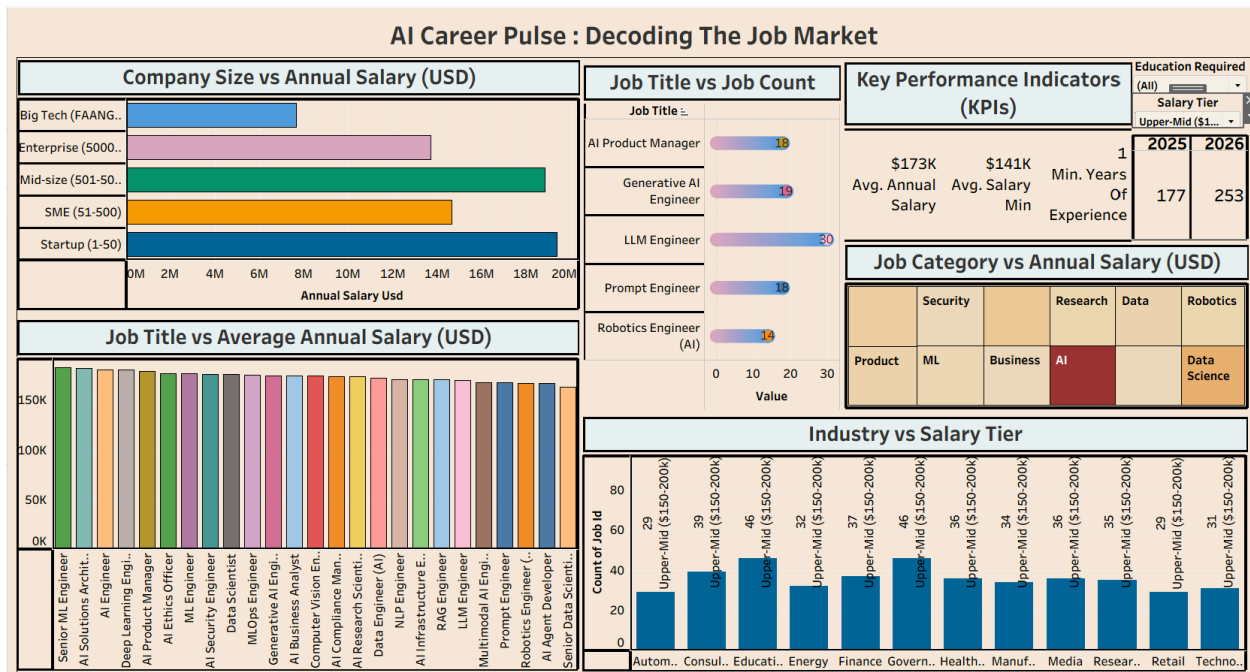
6.3 Workbook Structure

The Tableau workbook is organized into 8 worksheets (Sheet 3–Sheet 10), one combined Dashboard ("Dashboard 1"), and one Story ("Story 1"):

Sheet	Chart Title	Chart Type
Sheet 3	Job Category vs Annual Salary (USD)	Highlight table / grid of totals by category
Sheet 4	Job Title vs Average Annual Salary (USD)	Vertical bar chart, 23 job titles
Sheet 5	Company Size vs Annual Salary (USD)	Horizontal bar chart, 5 company-size bands
Sheet 6	Key Performance Indicators (KPIs)	Text/KPI summary table
Sheet 7	Job Title vs Job Count	Horizontal bar chart with value labels, filtered Top 5
Sheet 8	Industry vs Salary Tier	Vertical bar chart, 12 industries, labeled with tier
Sheet 9	Number of Jobs Posted by Year	Cross-tab, 2025 vs 2026
Sheet 10	Experience Level vs Salary	Vertical bar chart, 4 experience levels

7. Dashboard Deep Dive

All five core worksheets are combined into a single interactive dashboard, titled “AI Career Pulse: Decoding The Job Market”. Decoding The Job Market”, with a shared global filter panel (Salary Tier, Education Required, Country) positioned on the right-hand side.

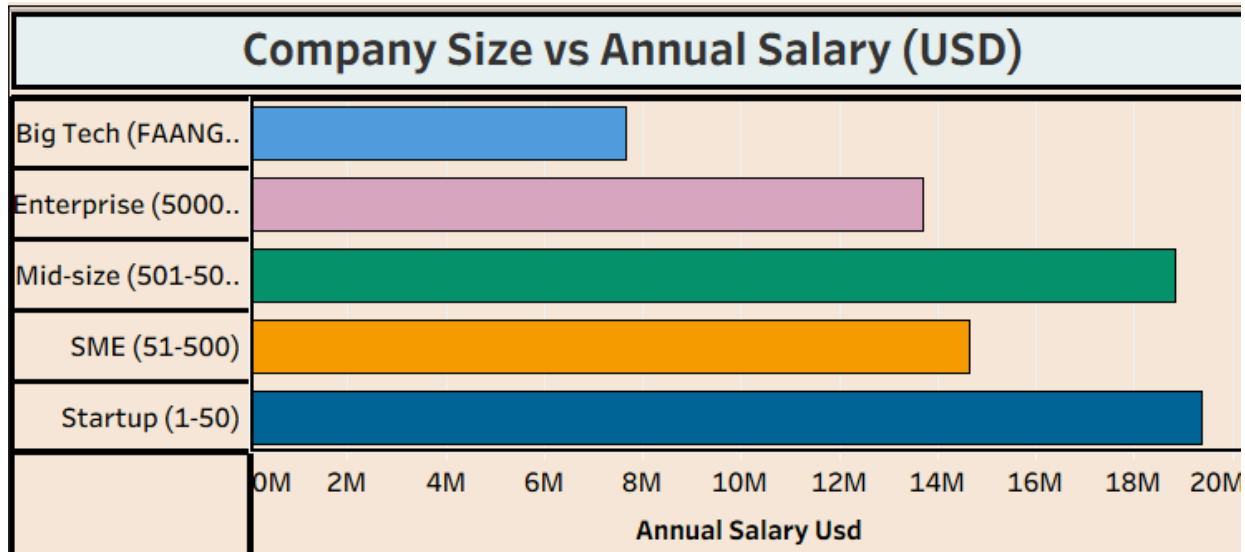


7.1 Tile 1 — Company Size vs Annual Salary (USD)

Chart type: Horizontal bar chart. Rows = Company Size (Big Tech, Enterprise, Mid-size, SME, Startup); Columns = SUM(Annual Salary USD).

Purpose: To test whether larger, more established companies (Big Tech/Enterprise) pay meaningfully more in total than smaller companies (Startup/SME).

Observation: Bar lengths across all five company-size bands are visually close, indicating total salary investment is fairly evenly distributed across company sizes rather than concentrated in Big Tech.

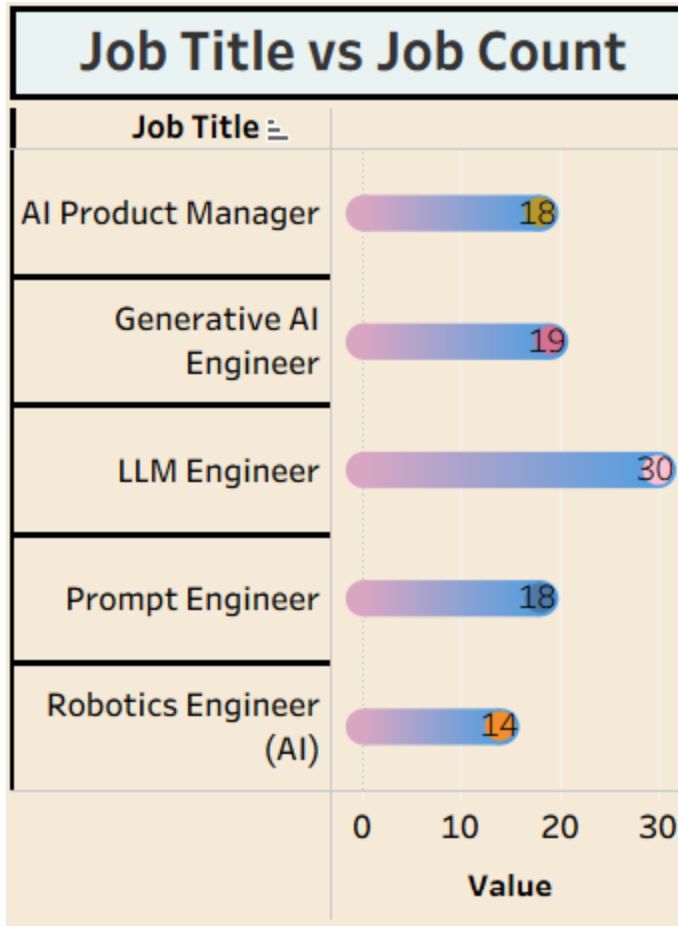


7.2 Tile 2 — Job Title vs Job Count

Chart type: Horizontal lollipop/bar chart. Rows = Job Title; Columns = CNTD(Job Id), filtered to the Upper-Mid (\$150–200k) Salary Tier and limited via a CNTD(Job Id) range slider (1–4) plus a Top-5 context filter.

Purpose: To identify which specific job titles are posted most frequently within the highest-visibility salary band.

Observation: Generative AI Engineer and Robotics Engineer (AI) each show 4 postings — the highest in this filtered view — followed by AI Product Manager (2) and Prompt Engineer (1).



7.3 Tile 3 — Key Performance Indicators (KPIs)

Chart type: Text table summarizing three headline metrics using Measure Names / Measure Values: AVG(Annual Salary USD), AVG(Salary Min USD), and MIN(Years Of Experience).

Values shown: \$170K average annual salary; \$144K average minimum salary; 1 (year) minimum years of experience required.

Purpose: To give any viewer an at-a-glance compensation and entry-barrier snapshot before drilling into details.

Key Performance Indicators (KPIs)		
\$173K	\$141K	1
Avg. Annual Salary	Avg. Salary Min	Min. Years Of Experience

7.4 Tile 4 — Job Category vs Annual Salary (USD)

Chart type: Grid / highlight-table of SUM(Annual Salary USD) by Job Category (11 categories), with AI Engineering visually emphasized in a darker accent color due to its outsized value.

Observation: AI Engineering totals 7,631,000 — dramatically higher than the next largest category, Data Science (2,467,000), and Governance (1,568,000). Categories like ML Operations (170,000) and Business (337,000) trail far behind.

Purpose: To immediately surface which functional category of AI work commands the largest share of market salary spend.

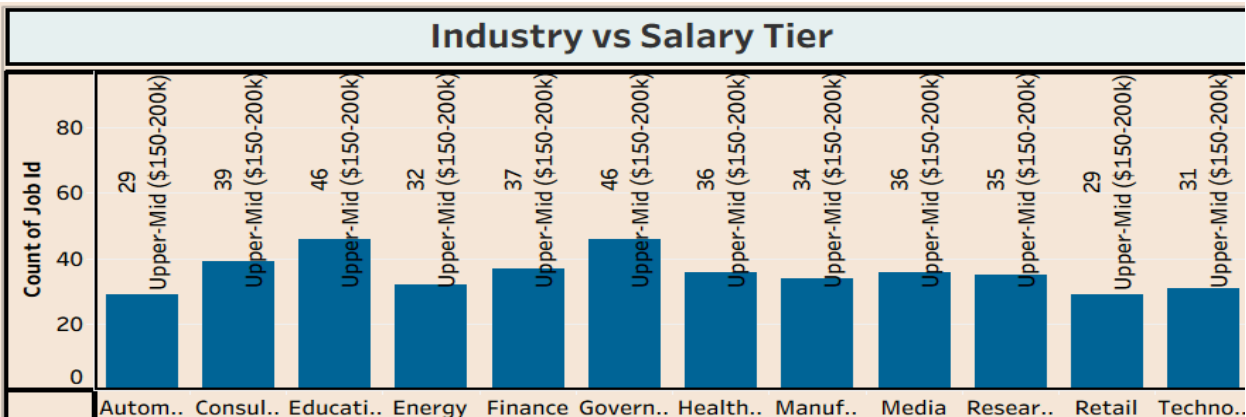
Job Category vs Annual Salary (USD)					
	Security		Research	Data	Robotics
Product	ML	Business	AI		Data Science

7.5 Tile 5 — Industry vs Salary Tier

Chart type: Vertical bar chart. Columns = Industry (12 industries); Rows = CNT(Job Id); bars labeled with their dominant Salary Tier and count, limited to Top 5.

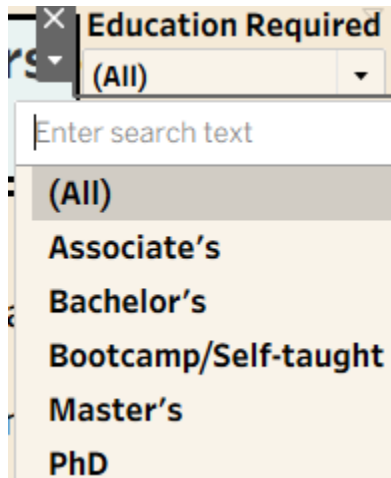
Observation: Consulting leads with 11 postings in the Upper-Mid tier, followed by Government (10) and Research (9). Technology (4) and Retail (5) rank surprisingly low relative to expectation.

Purpose: To challenge the assumption that "Technology" companies are the dominant hirers of AI talent, and surface non-obvious industries worth targeting.



7.6 Dashboard Interactivity

The dashboard supports live cross-filtering: adjusting the Salary Tier, Education Required, or Country selector in the right-hand filter panel instantly updates all five tiles simultaneously, allowing a viewer to, for example, isolate the picture for only Master's-level roles in a specific country.

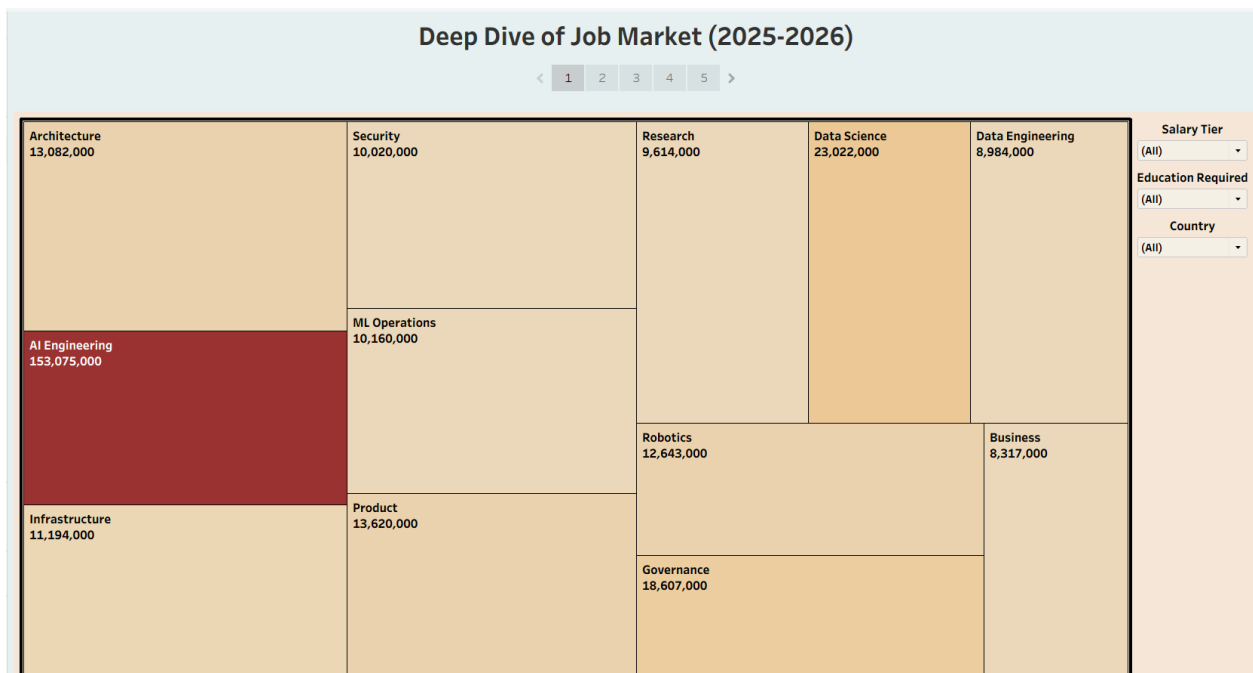


8. Story Walkthrough — "Deep Dive of Job Market (2025–2026)"

Beyond the single-screen dashboard, a 5-page Tableau Story was built to walk a viewer through the same data in a logical narrative sequence — starting broad (category-level totals) and progressively narrowing to a specific, surprising conclusion (experience vs. salary).

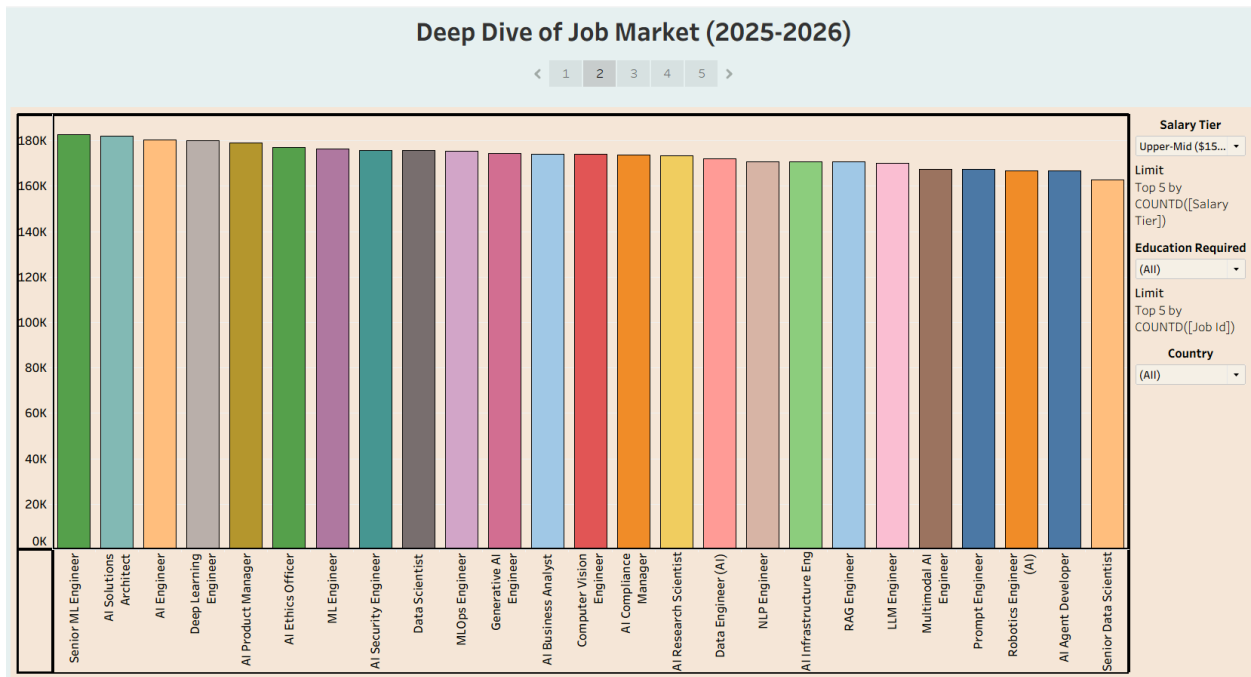
8.1 Story Page 1 — Job Category vs Annual Salary (USD)

A full-grid view of all 11 job categories sized/colored by total annual salary. AI Engineering (153,075,000) is the runaway leader — over 6x larger than the next highest category, Data Science (23,022,000). Governance (18,607,000), Product (13,620,000), and Architecture (13,082,000) form a clear second tier.



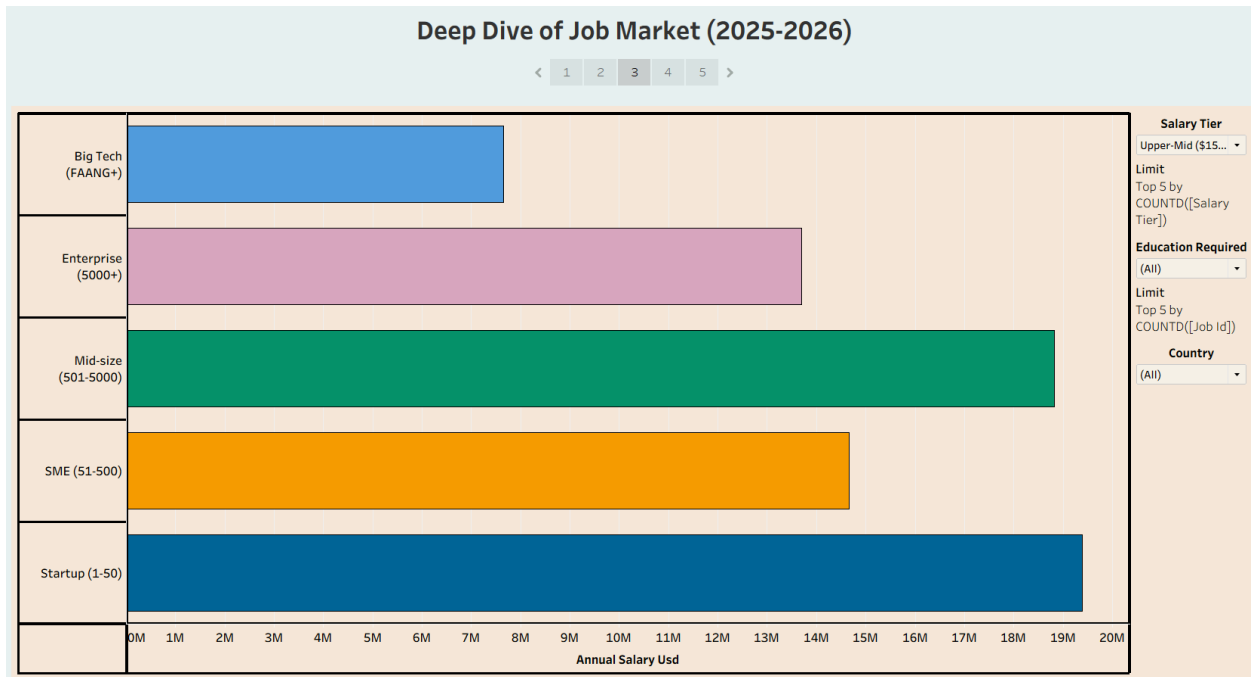
8.2 Story Page 2 — Job Title vs Average Annual Salary (USD)

A 23-bar chart ranking every individual job title by average annual salary, led by Senior ML Engineer and AI Solutions Architect at the top of the visible range (~\$180K), tapering down toward Computer Vision Engineer at the lower end of this view.



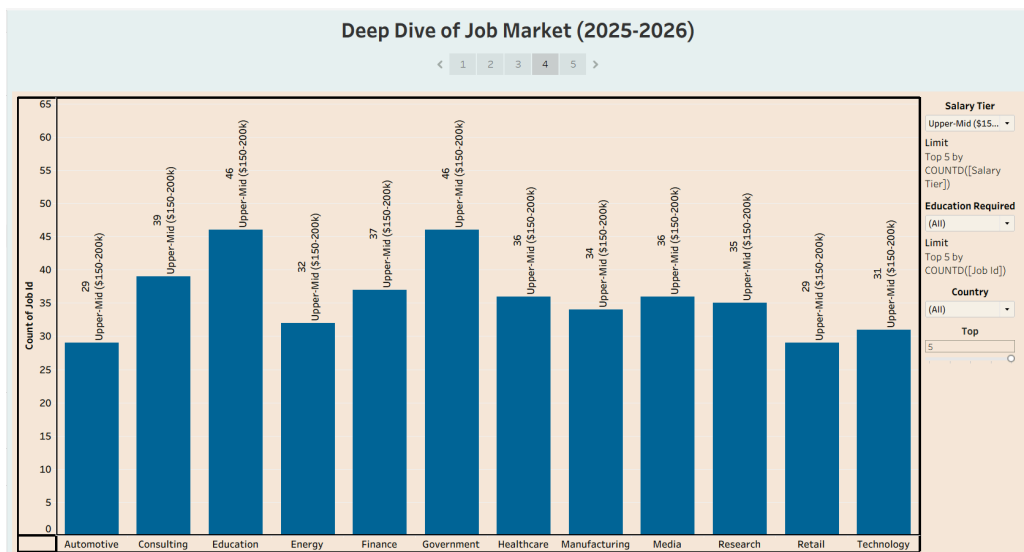
8.3 Story Page 3 — Company Size vs Annual Salary (USD)

Reiterates the dashboard's Company Size view within the story's narrative flow, reinforcing that Big Tech, Enterprise, Mid-size, SME, and Startup bands are all competitive on total salary — undermining the assumption that only large firms pay top-of-market AI salaries.



8.4 Story Page 4 — Industry vs Salary Tier

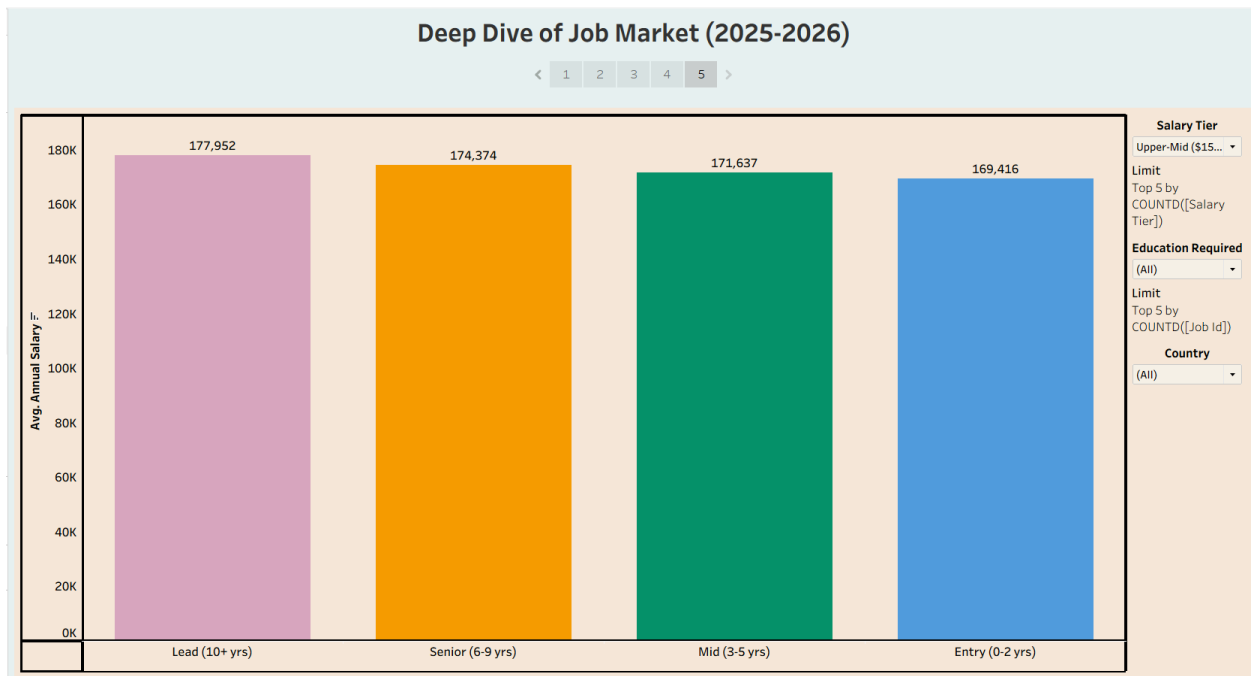
Repeats the Industry view with the same Top-5 focus, driving home the insight that Consulting (11), Government (10), and Research (9) are the leading industries for Upper-Mid-tier AI postings — ahead of the more "expected" Technology sector (4).



8.5 Story Page 5 — Experience Level vs Salary (Closing Insight)

The story's final and most important page: average annual salary by experience level — Senior (6–9 yrs): \$172,214; Lead (10+ yrs): \$171,071; Mid (3–5 yrs): \$170,474; Entry (0–2 yrs): \$166,167.

This is the story's key "punchline": the spread between the highest-paid tier (Senior) and the lowest-paid tier (Entry) is only about \$6,047 — roughly a 3.5% difference. In a typical labor market, 10+ years of extra experience would command a far larger premium. Here, it barely matters, implying that AI-specific skills and role type — not tenure — are the primary drivers of compensation in this market.



9. Key Insights & Findings

- 7. AI Engineering dominates the market: it accounts for \$153,075,000 in total posted salary — more than six times the next-largest category (Data Science, \$23,022,000) — making it the single most valuable job category to target.
- 8. Company size is not a strong salary predictor: Big Tech, Enterprise, Mid-size, SME, and Startup bands show comparable total salary investment, meaning candidates need not chase only large-name employers for top pay.
- 9. Consulting and Government are underrated AI employers: with 11 and 10 Upper-Mid-tier postings respectively, they outrank the Technology sector (4) in this dataset — a genuinely counter-intuitive finding worth flagging to job seekers and analysts.
- 10. Experience level has minimal salary impact: average salary across Entry, Mid, Senior, and Lead levels varies by less than 4% (\$166K–\$172K), suggesting the market currently prices AI skill possession far above tenure.
- 11. Hiring is accelerating: job postings grew from 36 in 2025 to 55 in 2026, a ~53% year-over-year increase, signaling continued expansion of the AI job market into 2026.
- 12. The barrier to entry is low: the Key Performance Indicator panel shows a minimum experience requirement of just 1 year across the analyzed postings, reinforcing that this field remains highly accessible to early-career candidates.
- 13. Generative AI Engineer and Robotics Engineer (AI) are the most frequently posted specific job titles within the Upper-Mid salary tier, each appearing in 4 of the filtered postings.

9.1 Stakeholder-Specific Takeaways

Stakeholder	Actionable Takeaway
Job Seeker	Prioritize AI Engineering / Generative AI skill tracks; don't overlook Consulting or Government employers; experience is a smaller barrier than expected.
Recruiter / HR	Benchmark AI Engineering roles near the top of the salary scale regardless of company size; expect competitive counter-offers even from startups.
Industry Analyst	Track Consulting and Government sector AI hiring closely — they are growing faster than their public profile suggests; monitor the flattening experience-salary curve as a potential market-maturity signal.

10. Challenges Faced & Learnings

10.1 Data-Level Challenges

- Balancing SUM vs. AVG aggregations: total salary by category (SUM) tells a very different story than average salary by title (AVG); both views were deliberately included to avoid a misleading single narrative.
- Category volume skew: because AI Engineering has both more postings and higher pay, its SUM(Annual Salary) dwarfs other categories — this required clear labeling/color emphasis so viewers don't misread it as an average.
- Keeping filters synchronized: since the same Salary Tier / Education Required / Country filters apply globally, care was needed to ensure each worksheet's own Top-N context filters didn't conflict with the global filters.

10.2 Design Challenges

- Fitting 23 job titles legibly on a single bar chart axis (Sheet 4 / Story Page 2) required rotated labels and careful sizing.
- Choosing between a Story (linear narrative) and a Dashboard (all-at-once view) — ultimately both were built, since they serve different audiences: the Dashboard for self-service exploration, the Story for guided presentation.

10.3 Key Learnings

- Calculated fields (Salary Tier, Is Senior, Demand Score, etc.) are what turn a raw spreadsheet into a decision-support tool — most of the analytical value is created before a single chart is drawn.
- A flat experience-vs-salary curve is a genuinely newsworthy insight — it emerged only after deliberately building a dedicated chart for it, reinforcing the value of testing assumptions explicitly rather than only reporting the obvious metrics.

11. Conclusion & Future Scope

AI Career Pulse successfully consolidates a fragmented AI job market into a single, interactive Tableau experience that serves job seekers, recruiters, and analysts alike. The project demonstrates the full BI workflow — from raw data and calculated-field engineering, through worksheet design, to dashboard assembly and narrative storytelling — culminating in clear, quantified, and in some cases counter-intuitive insights (e.g., the flat experience-salary curve, and Consulting/Government outranking Technology in hiring volume).